

Priyansh Garg

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EDUCATION

- 2026 – **Ph.D., Mechanical Engineering**
University of Minnesota, Twin Cities, MN, USA
- 2020 – 2024 **B.E., Manufacturing Engineering**
Birla Institute of Technology and Science, Pilani, Rajasthan, India
Thesis: Distributional Reinforcement Learning in a neural network model using dopaminergic neurons
Advisors: Sara Matias, Jay Hennig | *Co-Advisor:* Amit R. Singh

RESEARCH EXPERIENCE

- 2023 – 2024 **Bachelor's Thesis, Harvard University**, Cambridge, MA, USA
Supervisors: Dr. Sara Matias, Dr. Jay Hennig | *PI:* Dr. Naoshige Uchida
- Enhanced classical TD learning in an RNN model to dual learning rates for optimistic and pessimistic neurons by modifying the existing RNN code and adjusting it for episodic trials to enable the use of multiple cues.
 - Used a family of value predictors to find the best combination for reward distributions.
- Spring 2023 **Research Intern, IIT Delhi**, New Delhi, India
Supervisor: Dr. Saurabh Gandhi
- Analysed 128 Hz EEG neural datasets across three cortices in healthy and Alzheimer's cohorts.
 - Used current-based decomposition and an RNN network model to map connectivity alterations across distinct stimuli.
 - Evaluated regular, small-world, and scale-free network structures with varying degrees to identify optimal structural fits.
- Spring 2023 **Research Intern, CEERI Pilani**, Rajasthan, India
Supervisor: Dr. Dhiraj Sangwan
- Created a computer vision classification model to detect crop defects from seed X-ray scans.
 - Achieved 90% segmentation accuracy across real-time conveyor footage; engineered a functional UI interface using Qt.

TEACHING EXPERIENCE

- 2022 – 2024 **Undergraduate Teaching Assistant, Department of Mechanical Engineering, BITS Pilani**
Served as a teaching assistant under Dr Amit R Singh across multiple courses:
- Computer Aided Design (ME F318)* — Spring 2023, Spring 2024
 - Materials Science and Engineering (MF F216)* — Fall 2023
 - Machine Drawing (ME F217)* — Fall 2022

PROFESSIONAL EXPERIENCE

- 2024 – 2026 **Data Scientist, Hero FinCorp**, Gurugram, HR, India
- Developed an XGBoost model to predict customer delinquency risk; achieved AUC 0.89.
 - Engineered an event-driven pipeline via self-triggered Databricks jobs (Python/S3) to automate recurring ad-hoc requests, reducing manual effort by ~80%, saving ~15 hours weekly.

REFERENCES

Dr. Amit R. Singh, Assistant Professor, BITS Pilani, India

Dr. Jay Hennig, Assistant Professor, Baylor College of Medicine, USA

Dr. Sara Matias, Postdoctoral Research Associate, Harvard University, USA

Dr. Saurabh Gandhi, Assistant Professor, BITS Pilani, India